

2022

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer any **five** questions in which*

Q. No. 1 is compulsory.

1. Answer any **four** questions within **50** words each : 14

(a) Write down the Bessel's function of Zeroth order.

(b) Show for any function $F(t)$ and $L\{F(t)\} = f(s)$

$$L\{F'(t)\} = sf(s) - F(0)$$

(c) Define Fourier Cosine transform.

- (d) Define Geodesics.
- (e) Define eigenvalue and eigenfunction.
- (f) Give some advantages of C⁺⁺ programming over Fortran programming.

2. prove the following recursion relations of Legendre's polynomial $P_n(\mu)$: 7+7 = 14

(a) $n P_n(\mu) = (2n - 1) \mu P_{n-1}(\mu) - (n - 1) P_{n-2}$.

(b) $(\mu^2 - 1) \frac{d P_n(\mu)}{d\mu} = n \{ \mu P_n(\mu) - P_{n-1}(\mu) \}.$

3. Establish the following recursion or recurrence formula for Bessel's function of nth order ($J_n(x)$):

14

(a) $x J_n'(x) = n J_n(x) - x J_{n+1}(x)$

(b) $x J_n'(x) = -n J_n(x) + x J_{n-1}(x)$

4. Define Laplace Transform. Find the Laplace Transform of the functions : 14

(a) $F(t) = t^n$

(b) $F(t) = e^{\alpha t}$

5. (a) State and prove the first shifting theorem in Laplace Transforms. 10

(b) Slove : If $L\{\sin 3t\} = \frac{3}{s^2 + 9}$, then

$$L\{e^{2t} + \sin 3t\} = ? \quad 4$$

6. Discuss the determination of mass, radius, luminosity and temperature of a star. 14

7. Discuss the Gauss elimination method to find root of the functions. 14

8. Describe the solution of first order differential equation by Runge-Kutta method. 14

9. Write notes on any two of the following :

$$7 \times 2 = 14$$

(a) Hermite functions

(b) Age of the universe

(c) HR diagram

(d) Simpson's 1/3 rule

